HeatSafe Pakistan

AI-powered heat intelligence for safer outdoor work

HACKINSIGHT 2026 / PROJECT SUMMARY / 21 OCT 2026

4	4	210,432	21
evaluated ML cities	weather-feed cities	historical hourly rows	Prefect run records

THE CHALLENGE

Outdoor workers and site managers need clear heat-risk information to plan precautions. HeatSafe combines historical forecasting evidence, live weather monitoring and cited workplace-safety guidance.

THE SOLUTION

01 DATA
ERA5 weather, quality rules, and DuckDB evidence.

02 PREDICT
City-specific heat-index ML, evaluated against baselines.

03 EXPLAIN
Gemini RAG using NIOSH, HSE and NOAA sources.

2025 HELD-OUT TEST RESULTS

24-hour-ahead weather-derived heat index. Lower MAE is better.

CITY	MODEL MAE	BASELINE MAE	RESULT
Karachi	1.056°C	1.107°C	Improved
Lahore	1.875°C	1.948°C	Improved
Islamabad	1.798°C	1.852°C	Improved
Multan	1.638°C	1.713°C	Improved

LIVE COVERAGE & AUTOMATION

KARACHI - LIVE ML

Prefect hourly workflow updates a DuckDB gold-table forecast. Scheduled completed runs were verified.

FOUR-CITY WEATHER FEED

Provider-issued weather forecasts for Karachi, Lahore, Islamabad and Multan; not ML predictions.

IMPORTANT LIMITATIONS

Four-city ML scores are historical. Live ML currently serves Karachi only; full DQ processing for added cities is pending. Heat index is not worksite WBGT or a safety guarantee.

SOURCE EVIDENCE ERA5 / Open-Meteo • modeling/evaluations/ • evidence/prefect_runs.csv

github.com/Hamdanilyas01/Hack_insight-